Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Distributed, AI-driven Robotic Interiors to Promote Routine Anchoring for the Elderly through Personalized Environmental Reconfigurations*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio · Medical		*Mechanical · Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Functional Restoration / Assistive & Welfare Devices	*Cognitive / Sensory Function Support Devices*	*Robotics/ Automation Machinery*	*Robot Design Technologies*
Concept of Proposed Technology	***Concept:*** *"Robotic Interiors" are AI-driven robots that are portable, distributed, and seamlessly embedded in everyday interior spaces. We propose to design and develop a set of robotic interiors for senior adults' homes and nursing homes: the robotic interiors can reconfigure the light, sound, and movement of interior environments to anchor the daily routines of senior adults, especially those*			

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	with MCI (Mild Cognitive Impairment) and dementia. Interior environments can anchor daily routines because human cognition and activity are tied to places/environments rather than instructions [1]. This relationship is known as Activity–Environment Coupling. ***Rationale:*** *Cognitive decline is a common phenomenon and serious issue for senior adults: For people aged 65 or older in South Korea, 8.1% (878000) have dementia and 24.1% (2.6 million) have MCI [2]; In the US, 10% (6.12 million) have dementia and 22% (13.5 million) have MCI [9]; Worldwide, the number of people with dementia alone is expected to double every 20 years, reaching 78 million in 2030 and 139 million in 2050 [10]. People with MCI and dementia face significant safety risks and emotional distress at home due to cognitive decline, including difficulties managing medications, forgetting to turn off appliances, uncertainty about what to do next, disoriented wandering, and a loss of daily control that leads to emotional insecurity and frustration.* *"Routine anchoring" is the practice of establishing consistent, predictable daily patterns and activities that provide a sense of stability, security, and comfort for the elderly, particularly those with cognitive decline, such as MCI and dementia. Routine anchoring is crucial for people with cognitive decline because it externalizes memory, reduces cognitive and emotional strain, and sustains autonomy and identity. Routines and memories are encoded with spatial associations — people recall and enact activities in relation to where they occur [3, 4]. This is especially true for elders, whose long-established activity patterns are strongly tied to places within their home or nursing home (e.g., favorite chair, cozy reading corner, kitchen table, nighttime path to bathroom, etc.). Thus, we propose that older adults' living environments should function as embodied, spatial cues—provided at the right time and in the right way—to support memory and action through predictable spatiotemporal patterns rather than explicit instruction or surveillance. To best support seniors, their living places need to be responsive, adaptive, and intelligent by recognizing seniors' behavioral patterns and customizing environmental cues based on individual habits. Thus, "Robotic Interiors" is the key concept of the proposed technology.* ***Key features:*** 1) ***Hardware: Distributed, Portable, and Discreet.*** *The robotic interior components will be small-scale, portable, and seamlessly integrated into seniors' living places (e.g., home or nursing homes) in a natural, non-intrusive way.* 2) ***Software: AI-powered Personalization.*** *The robotic interior prototypes will not only allow manual setup (e.g., via buttons or knobs) but also employ vision-language-based or sensor-based Human Activity Recognition (HAR) algorithms [7, 8] to identify user behavioral patterns, enabling robotic components to further adjust environmental cues based on individual living habits. Notably, user feedback during the human-centered design process will dictate whether the HAR algorithm is sensor-based or vision-language-based.* 3) ***Interaction: Human-centered Design.*** *The robotic interior prototypes will be developed through an iterative, human-centered design and engineering process that carefully considers user perceptions, preferences, emotions, and well-being.*
Rationale for Joint R&D	*The proposed project requires close collaboration between a research institute and an industrial*

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Project	*partner to successfully translate robotic interior prototypes into a practical, market-ready home robotics system. A joint R&D framework is essential to integrate HAR algorithms with robotic hardware and to validate the technology under real-world conditions.* *Rationale for collaboration:* *1) The research institute provides expertise in human-centered product design and evaluation, hardware development (including both material fabrication and electronics engineering), and system integration for robotic interior prototypes.* *2) The Korean SME partner will make efforts in applying the existing vision-language-based or sensor-based HAR algorithms [7, 8] for the context and purpose of this project (i.e., recognizing senior adults' behavioral patterns at home and nursing homes). Notably, the research institute can provide technical assistance if needed.* *3) The Korean SME partner provides expertise in market research, product manufacturing, and product commercialization, enabling a smooth transition from in-lab software and hardware research (e.g., HAR algorithm adoption and robotic hardware prototyping) to market-ready or even industry-level products.* *4) Effective co-design of AI algorithms and robotic hardware is necessary throughout the whole design research process to meet cost, safety, and reliability constraints.* *Expected benefits of joint R&D:* *1) Accelerated transition from partially functional robot prototypes in the lab to deployable, AI-powered robot prototypes testable in the field.* *2) Reduced technical risk and improved user satisfaction through iterative, human-centered hardware–software co-design and validation.* *3) Creation of a solid, scalable foundation for the commercialization of distributed, AI-driven robotic interiors for the elderly.*
Key Words	1. *Robotic Interiors* 2. *AI-powered Personalization* 3. *Multimodal Environmental Cues* 4. *Routine Anchoring for the Elderly* 5. *Activity–Environment Coupling*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2)]	**Start of the Project**	Level (2)	**End of the Project**	Level (6)

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R&D Objectives	*The objective of this R&D project is to develop a set of distributed, AI-driven robotic interiors that can intelligently and reliably reconfigure interior environments to promote routine anchoring for the elderly. The specific R&D objectives are:* • *Design distributed robotic components (e.g., robotic handles, decorations, hinges, drawers, nightlight, wall-mounted storage) and understand how these components will reconfigure environments as well as how human-robot interactions may unfold.* • *Develop the hardware for the robotic interior components, including the mechanical mechanism, physical embodiment, and electronic circuits.* • *Apply vision-language-based or sensor-based HAR algorithm [7, 8] to the context (i.e., home or nursing home) and purpose (promote senior adults' routine anchoring) of this project.* • *Integrate the software (vision-language-based or sensor-based HAR algorithm) and hardware (mentioned above) into testable robot prototypes in real-world contexts.*
R&D Target Specifications	**Human Activity Recognition Performance (85%):** *Achieve the successful detection and classification of seniors' activities through vision-language-based or sensor-based perception with a successful rate of at least 85%.* **Usability Performance (excellent usability: SUS score of 80 or above)** *Achieve excellent usability for the final high-fidelity robot prototypes with the SUS (System Usability Scale [5]) score of 80 or above.* **Routine Anchoring Effectiveness (Qualitative and Quantitative Evaluation)** *Quantitative Evaluation: We will measure changes in personal habits and routines (e.g., taking medications at certain times, using the same nighttime path to the bathroom, morning reading in favorite chairs, etc.) using the Self-Report Habit Index (SRHI) [6] through before-and-after quasi-experiments. We anticipate that senior adults are better anchored (i.e., selected habit becomes more consistent, automatic, and ingrained [6]) after the distributed robotic interiors implemented in interior living environments. We are targeting a statistically significant p-value (<0.05).* *Qualitative Evaluation: We will conduct in-depth interviews with senior participants to explore the effectiveness of the robotic interior prototypes for routine anchoring. This qualitative evaluation will also provide further guidance on the final prototype improvement before the commercialization of market-ready products.*
R&D Plan	*This R&D project will be conducted over three years and will follow an iterative human-centered design, development, and validation strategy that moves from in-lab usability/performance testing to real-world summative evaluations at seniors' homes or nursing homes.* **Year 1: Generating Design Guidelines, Developing Low-fidelity Prototypes, and Exploring HAR Algorithms** 1. *Conduct customer discovery studies and market research to better identify and define the potential product markets.* 2. *Conduct in-depth interviews with senior adults, caregivers, and healthcare professionals to understand the everyday challenges of senior adults due to cognitive decline and memory loss, which will inform our understanding on which daily routines are critical yet easily*

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	forgotten for senior adults. 3. *Based on the interview and customer discovery study results, design guidelines will be generated.* 4. *Conduct co-design workshops with senior adults, caregivers, and healthcare professionals to generate specific design ideas within the concept of distributed robotic interiors for routine anchoring. These ideas will be developed into low-fidelity prototypes, which will focus on form factors and mechanical design of the robotic interiors.* 5. *Exploring different existing vision-language-based or sensor-based HAR algorithms [7, 8] and different electronic hardware (e.g., cameras, sensors, and processors) to see which software-hardware combinations can be the best fits for this product.* **Year 2: Robotic Prototype Development and Laboratory-scale Integration of HAR Algorithms for In-lab Formative Testing** 6. *Integrate sensors, actuators, and other required electronics into the low-fidelity prototypes to generate robotic prototypes.* 7. *Apply the selected HAR algorithms (based on the explorations in Year 1) into robotic prototypes.* 8. *Improve the HAR algorithm through iterative in-lab testing of the robotic prototypes.* 9. *Test the usability and UX for the robotic prototypes.* 10. *Develop iteratively the robot hardware and software with the standard of manufacturable, cost-efficient, and reliable systems in mind, laying a solid foundation for market-oriented, high-fidelity prototype in Year 3.* **Year 3: Develop High-fidelity Prototypes for Efficacy and Usability Evaluation in Real-world Settings** 11. *Develop high-fidelity robotic interior prototypes to address the usability and UX issues identified in Year 2 with commercialization and manufacturing requirements in mind.* 12. *Deploy the high-fidelity prototypes in realistic home or nursing home environments with senior adults (especially those who experience Mild Cognitive Impairment (MCI) or early Dementia) and evaluate the usability and effectiveness of this technology.* 13. *Refine system reliability, stability, and usability based on the field study results.* 14. *Assess commercialization potential and prepare the technology for industrial transfer.*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*The Research Institute will be responsible for the human-centered design and development of robotic hardware, technical support for applying HAR algorithms (if needed), and integration of the hardware and software in high-fidelity robot prototypes for efficacy and usability evaluations in real-world settings.* *The specific roles and responsibilities of the research institute include:*

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	1. *Conduct human-centered design and research activities (i.e., initial interviews, co-design workshops, and in-lab usability evaluations) to specify user needs, enact design guidelines, and keep users involved in the design and development process.* 2. *Provide technical support to the Korean SME partners for applying existing HAR algorithms to the context and purpose of this project, if needed.* 3. *Develop the hardware components (materials, mechanical mechanisms, and electronics) from low-fidelity prototypes to robotic prototypes to high-fidelity prototypes.* 4. *Integrate software and hardware components into the final high-fidelity prototypes.* 5. *Conduct real-world experiments with the high-fidelity prototypes, including experiment design, experiment conduction, and data analysis.* 6. *Provide technical support and collaboration with SME partner for system integration and experimental evaluation.*
Expected R&R of Korean SME Partner	*The Korean SME partners will play a key role in market research and translating the in-lab robot prototypes into practical, deployable, and reliable robotic products (including both hardware and software) as solid foundations for future commercialization.* *The specific roles and responsibilities of the Korean SME institute include:* 1. *Conduct customer discovery studies and market research to better identify and define the potential product markets in South Korea, the United States, and worldwide.* 2. *Explore existing HAR algorithms [7, 8] and apply the HAR algorithm with corresponding electronic components to achieve successful recognition of seniors' behavioral patterns in the contexts of home and nursing homes (with technical support from the research institute, if needed).* 3. *Develop the hardware and software of robotic and high-fidelity prototypes in the lab into manufacturable, cost-efficient, and reliable systems through Design for Manufacturing (DFM) and Design for Assembly (DFA), providing solid foundations for future commercialization.* 4. *Evaluate the production and commercialization potential of the high-fidelity prototypes for the finalized materials, components, and systems.* 5. *Provide technical support and collaboration with the research institute for system integration and experimental evaluation.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The proposed joint R&D project is expected to deliver a set of AI-driven robot prototypes that are portable, distributed, and integrated seamlessly into senior adults' living environments, either at home or nursing home. These robotic interior prototypes can reconfigure the light, sound, and movement of interior environments to anchor the daily routines of senior adults, especially those with MCI (Mild Cognitive Impairment) and dementia. Thus, this project investigates how smart rooms (or interior spaces) may partially play the supportive role of a caregiver for senior adults through constant and non-intrusive social presence, which is a topic at the very frontier of Interaction Design (ID), Human-Robot Interaction (HRI), and AI Ethics research.* *In addition, vision- or sensor-based HAR algorithms can provide robust activity identification for older*

	adults, benefiting not only this project but also broader applications in healthcare, education, human–computer interaction (HCI) [7], and other domains [7] with strong demand in the Korean, U.S., and international markets. *Finally, this project will provide the Korean SME partner with high-fidelity robotic interior prototypes whose hardware and software components are manufacturable, cost-efficient, and reliable. This will become a solid foundation supporting further technology commercialization to enable industry-level robotic solutions for homes and nursing homes. This technology will not only support safer and more independent living of senior adults but also reduce healthcare professionals' workload and mitigate the population aging crisis in South Korea, the US, and many other countries. Finally, this project is expected to strengthen the competitiveness of Korean robotics SMEs and contribute to the growth of the AI-driven service robot industry through international collaboration.*

6. References

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☐ Terms:

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1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization